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Identifying post-Covid Beveridge curve shifts in the Eurozone: a country-level analysis

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1 Introduction

The Beveridge curve (BC), which describes the inverse relationship between labour demand and supply as measured by vacancies and unemployment in the economy, has been a key tool in the analysis of the labour market since the middle of the last century. Originally identified by William Beveridge in 1944, this relationship has been the subject of scrutiny in academic works and policy proposals. Following the work of Pissarides, Diamond et al in the late 1980s, economists acquired the necessary tools to describe this relationship in more detail, while simultaneous advancements in econometric methods allowed researchers to express the movements of the curve in closed form. While a common approach in the literature is to test the Beveridge curve by estimating the matching function (Barnichon & Figura, 2015; Blanchard & Diamond, 1989), this work focuses on the short-term dynamics, or shifts, resulting from a large crisis such as Covid.

This paper addresses several questions about the Beveridge curve and the state of European labour markets between 2010 and 2024. While most attention is devoted to the immediate effects of the coronavirus, the analysis extends to the longer period from 2020 until today. General features of the workforce in Germany, Italy, and the UK in recent years are also described, providing deeper insight into how the tightness indicator for these countries may evolve in the future. For most countries in the study, the validity of the BC between 2010 and 2025 is affirmed. Hence, this work contributes to economic research on the BC in multiple ways:

1. It tests whether and how the BC held for the Eurozone in the past 15 years, in particular for Germany, the UK, and Italy.
2. It models the shift in the empirical BC for all three countries in the aftermath of the Covid crisis in order to quantify Covid's immediate impact on their labour dynamics.
3. It provides general features of the workforce and labour conditions in the EU, as well as in the countries in the sample, tying the effects of Covid with earlier trends to predict further developments in the state of the region's labour market, including its tightness.

The idea of testing for shifts in the BC following a crisis using time-varying dummies is a common tool in macroeconomic analysis and was used extensively after the Great Financial Crisis to describe its adverse effects on aggregate hiring (Daly et al., 2012; M. W. Elsby et al., 2010). For Covid, however, the literature is still scarce and is mainly focused on the labour market in the US. Other research on tightness in the Eurozone is primarily concerned with drawing more general conclusions about the state of the EU labour market by testing a model of the Beveridge curve on a panel of countries. This work, on the other hand, while addressing labour dynamics in Europe overall, takes a closer look at developments in specific countries.

The paper is organized into three sections. The first chapter reviews the economic theory underlying the Beveridge curve and the existing econometric methods to model this relationship using the available data. The second section provides an overview of the job market and labour force in the EU, with an emphasis on Germany, the UK, and Italy. It also discusses how Covid affected employment, participation, and hiring across different sectors and social groups, while summarizing the policies adopted by national governments in response to the shock. The section concludes with a visual representation of the Beveridge curves for all three countries, illustrating the changes in vacancies, unemployment, and the vacancy-to-unemployment ($v - u$) ratio before, during, and after Covid. The third and final chapter presents the model of the BCs for the three countries, which tests for their shifts in the wake of the pandemic. The results obtained for Germany, the UK, and Italy are then compared with one another, and the paper ends with a discussion of the implications of these findings for the labour dynamics in these countries.

2 A review of theory

2.1 The empirical relationship between vacancies and unemployment rates

The parting point for the BC was the original research by the British economist William Beveridge. Contrary to the contemporary view, Beveridge defined full employment not as zero unemployment, but as a situation where there are always more vacant jobs than unemployed people, laying the foundation for further scientific discussion on the topic (Beveridge, 1870). Although the "curve" was not mentioned explicitly, the idea that unemployment was a result of bargaining in the labour market, rather than a purely policy-induced phenomenon, was new.

Graphically, this relationship was first plotted for UK, using 1946-1956 data (Figure 1).

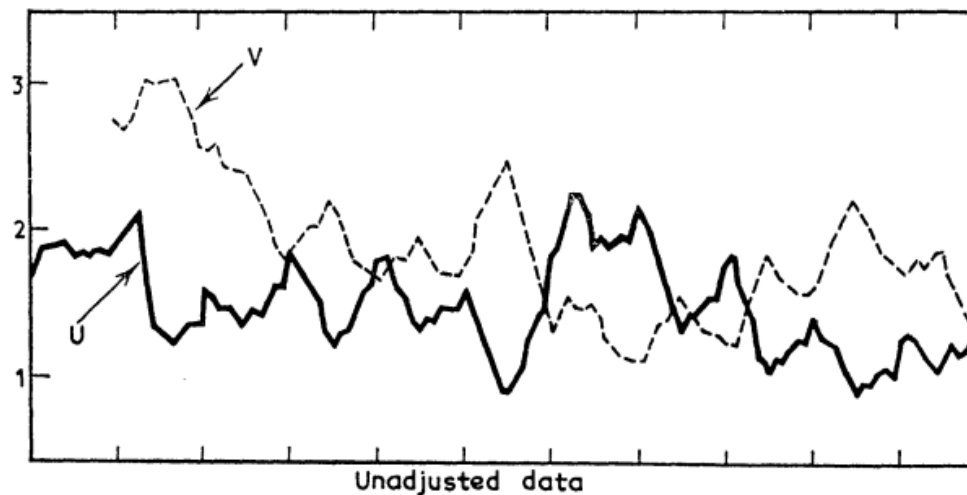


Figure 1: Vacancy and unemployment series for UK, 1946-1956. Source: Dow and Dicks-Mireaux (1958)

Furthermore, Dow and Dicks-Mireaux (1958) stipulated that the relationship was not only negative, but also convex. Between 1940s and 1970s, the inverse relationship between jobs and slack proved to hold well for historical datasets spanning UK, US and much of West Europe (Layard et al., 2005; ML, 1970). However, a dynamic framework to estimate labour market developments was lacking, and the empirical results were viewed more as a rule of thumb. Explanations were derived intuitively, but key factors were already predicted before a comprehensive theory was developed. Terms such as "demand pressure" (labour demand influencing flows into employment), "malalignment of jobs" (mismatch due to frictions caused by the intensity of search, skill gaps and other factors) appeared as early as in '50s (Dow & Dicks-Mireaux, 1958). In spite of this, the relationship was analyzed only in a static framework, as was typical for the mainstream economic dogma of the time. Much of neoclassical analysis, including the works of Cohen and Solow between '60s and '70s, relied on comparative statics (Rodenburg et al., 2010). Most work upon which current theory rests did not begin until the late '80s and early '90s.

2.2 Job search and the matching function

Novel developments in theory came independently from the UK (Mortensen, 1986; Pissarides, 2000) and the US (Blanchard & Diamond, 1989; Diamond, 1982) in the final quarter of the 21st century. The results of their joint efforts became known as the Diamond-Mortensen-Pissarides (DMP) model, which is briefly discussed below. The starting point in the DMP framework is that hiring may be represented by a function

of vacancies and unemployment, which capture trends in labour demand and supply (Pissarides, 2000). To this end, a matching function, assumed to have constant returns to scale, is used to describe the efficiency with which supply and demand interact in the labour market (Di Giorgio & Giannini, 2012):

$$h_t = \alpha \cdot m(u_t, v_t), \quad (1)$$

$$\frac{\partial m}{\partial u} > 0, \quad \frac{\partial m}{\partial v} > 0$$

Here, h_t denotes the number of hires at time t , u_t is the unemployment rate, and v_t is the job vacancy rate. The function $m(u, v)$ captures the matching process between unemployed individuals and job openings, and α is a parameter reflecting the overall efficiency of this process. The key assumption is that m is homogeneous, so that it can be expressed as a function of the ratio of v to u ¹.

Under these conditions, the Beveridge curve is typically convex to the origin and downward sloping in (u, v) space. This curve reflects a steady-state relationship, where changes in hiring efficiency or labor market dynamics can cause shifts in its position.

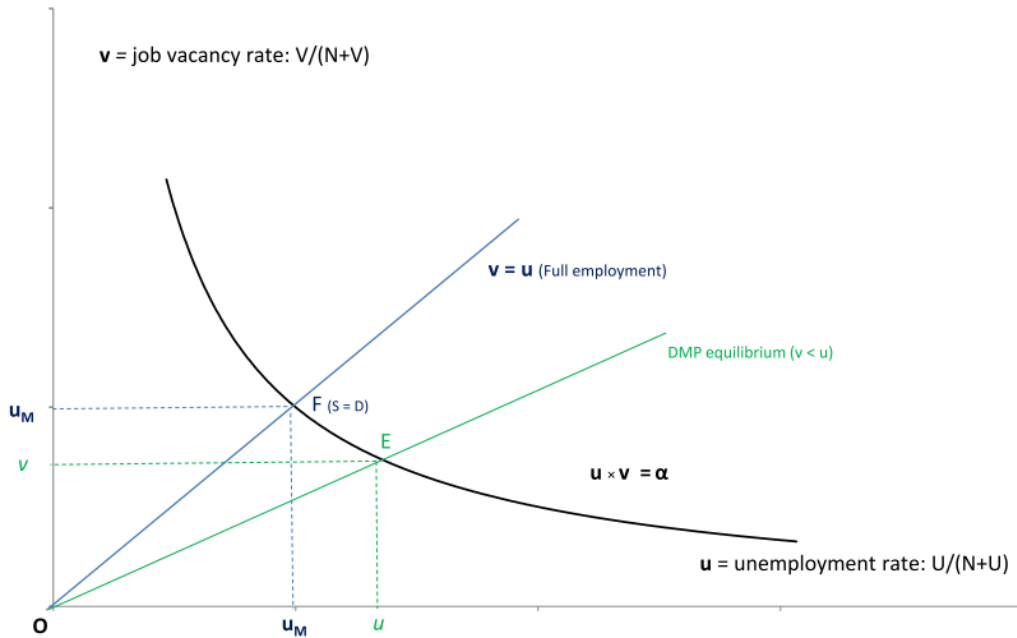


Figure 2: The Beveridge curve as a function in (u, v, λ) space, where λ is an exogenous rate of job loss. Source: Leythienne (2024)

For the scope of this research, it is sufficient to understand the intuition behind movements versus shifts in the curve. A good summary is given in Barlevy et al. (2024). The Beveridge curve is given by all pairs (v, u) for which the flow into unemployment equals the flow out of it (Pissarides, 2000):

$$\lambda(1 - u) = q(\theta)v \quad (2)$$

where λ is the job-separation rate and $q(\theta)$ - the job-finding rate (a function of labor market tightness $\theta = \frac{v}{u}$).

¹Petrongolo and Pissarides (2001) showed that the empirical matching function exhibited approximately constant returns to scale for US and a number of European countries.

Equation 2 describes a steady-state equilibrium, where mean unemployment remains unchanged. The curve is convex because, for an exogenously defined λ (both job-matching and job-separation are assumed to be processes following a Poisson distribution), q will respond negatively to higher θ , reflecting a decreased chance of being selected for a match from the pool of unemployed when potential workers are scarce relative to open positions. Assuming a tightening, two cases are possible: either v increases, driving θ upwards and $q(\theta)$ downwards until equilibrium is attained again at the steeper segment of the BC, or u decreases in a way that the left-hand side of the equation - mean quits - grow, reducing the pool of unemployed such that a higher θ will trigger less hires (right-hand side). Naturally, the same mechanism operates in reverse when the market becomes slack. The most intuitive way to explain the shape of the curve is by considering matching as a two-way bargaining process between employers and candidates: the higher the unemployment, u , given a fixed number of openings, v , the easier it is for a firm to hire. A small change in v will have a significant impact on u when the latter is relatively large. The opposite case takes place when there is a relative shortage of labour supply (u), and the slope is steeper.

An additional component of Figure 2 is the so-called job creation curve (JCC), which is derived from equilibrium wage equations (Pissarides, 2000). The JCC reflects the demand side of the labour market, incorporating structural frictions (e.g., skill mismatches), the effects of unemployment policies, and the broader state of the economy. The intersection of the BC with the JCC at a given point in time gives the θ - labour tightness. The derivation stems from treating jobs, job search and vacancies as productive assets generating utility for workers and firms (M. W. Elsby et al., 2015):

$$r\mathcal{U} = b + f(\theta)(\mathcal{W} - \mathcal{U}) \quad (3)$$

$$r\mathcal{W} = w + \lambda(\mathcal{U} - \mathcal{W}) \quad (4)$$

$$r\mathcal{V} = -pc + q(\theta)(\mathcal{J} - \mathcal{V}) \quad (5)$$

$$r\mathcal{J} = p - w - \lambda\mathcal{J} \quad (6)$$

The first two equations describe the discounted value of unemployment ($\mathcal{U}$) and work ($\mathcal{W}$), respectively, from the point of view of a worker, while the latter two characterise the valuation of vacant ($\mathcal{V}$) and filled ($\mathcal{J}$) openings for a firm. Here, b is the parameter capturing all the benefits that an unemployed person gets during this state (e.g. unemployment insurance), w denotes the wage, c is the cost incurred by the firm for maintaining a job opening and p measures the productivity of labour. Imposing the requirement that $\mathcal{V} = 0$ in equilibrium (the firm opens vacancies until the present value of an additional opening is driven to zero) and assuming that the surplus from vacancy-filling is split according to Nash bargaining, the job creation condition is given by:

$$(1 - \beta)(p - b) = pc \left(\beta\theta + \frac{(r + \lambda)}{q(\theta)} \right), \quad (7)$$

Implying that θ is independent of unemployment: p , b and c are parameters denoting labour productivity, benefits to the worker from remaining unemployed (due to unemployment insurance, extra leisure, alternative activities etc.) and the cost of hiring for the firm, respectively; β indexes the bargaining power of workers. Therefore the equilibrium depends on two processes: flow dynamics, captured by (2), and wage-setting and labour demand, stemming from (3)-(7). Graphically, we get that market tightness is found at the intersection of the BC (flow) and JCC (labour demand):

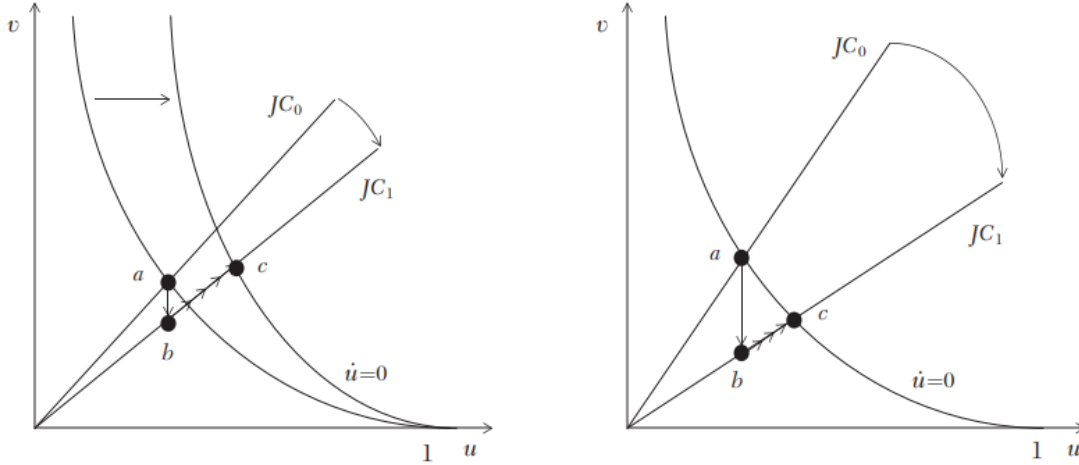


Figure 3: Labour dynamics through the lens of BC and JC. Source: M. W. Elsby et al. (2015)

The first image in represents a situation where unemployment increases due to a shock to λ or $m(uL, vL)$, leading first to an outward shift of the BC and then, by disintivating vacancy posting due to either lower present value of filled jobs (eq. (5) and (6), when λ falls) or lower q (when m deteriorates), to the clockwise rotation of the curve². The second image depicts a situation where only labour demand falls, without a shift in the BC (suppose when p decreases), which causes a clockwise rotation of the JC and a lower value of θ .

The original model was a big step from static to dynamic modelling of the labour market, where flows into and out of the labour force are explicitly accounted for. Its largest drawback is that persistent shifts in BC observed in major Western economies can be explained only by the deterioration in matching technology (M. W. Elsby et al., 2015). Moreover, it does not consider that job search happens not only among the unemployed, but among the employees, too, and that transitions from inactivity to unemployment and vice-versa play an important role in defining the equilibrium condition. Recent evidence suggests that these factors are not only significant in explaining the tightness of the market, but perhaps even more so than traditional unemployment-employment dynamics (Fujita, 2010; Fujita & Moscarini, 2017). Other authors found that some parameters explain actual employment dynamics better or worse than predicted by the DMP model (Mortensen & Nagypal, 2007; Shimer, 2005). We do not stop on further developments in economic theory regarding job search and matching as it goes beyond the scope of this paper. Instead, we argue that the original theory had major implications for econometric analysis, as it provided a "legal" economic basis for policy practitioners, who could now fully exploit the BC to describe the state of labour markets.

2.3 From economic to econometric modeling

As befits economics, quantifying important theoretical findings is not easy. The $v - u$ relationship is not an exception, especially when it comes to dynamic analysis. One of the earliest works in this direction was done by Blanchard (1989), who estimated aggregate hiring (H) in US as a function of lags of vacancy (V) and unemployment (U) stocks at time t . The basic specification took the form:

$$\ln(H_t) = a + b \cdot t + c \ln(V_{t-1}) + d \ln(U_t) + \epsilon_t \quad (8)$$

²In case of shocks to λ , the JCC may also rotate counterclockwise, leading to more vacancies and higher equilibrium tightness: since job-exit rate is higher, so will be the stock of unemployed. By the properties of m and (5), the expected return from opening a vacancy will rise accordingly. If this positive effect prevails, then a higher λ will lead to more vacancies per each level of u , rotating the JCC downwards.

where bt captures a linear trend and c and d summed to 1 to satisfy the homogeneity assumption. The log specification is common tool to express the BC algebraically, used extensively for estimating the matching function parameters and the effects of unemployment and vacancy dynamics on aggregate matching. The choice of the dependent log-variable depends on the scope of the research, but it is of most use when either the elasticity parameter α or the matching efficiency parameter σ are tested (M. W. Elsby et al., 2015; Gottfries & Stadin, 2025; Kangasharju, 2005). It may also be used for historical analysis of the $v - u$ relationship itself. To identify the drivers of gradual shifts in the BC over time, Blanchard (1989) relies on instrumental variables. Decomposition of changes in employment pool according to the origins of the workforce (i.e., unemployment or inactivity) is another tool explored by econometric literature to characterize structural conditions of labour (Yashiv, 2006). There is a large body of research where labour dynamics are studied by estimating the behavior of a matching function (Barnichon & Figura, 2015; Gottfries & Stadin, 2025; Yashiv, 2006). A recent extension of this approach is to estimate a level of efficient employment in the economy, where the mismatch unemployment is measured as an excess of unemployment over the value of vacancy rates. This framework looks at total amount of vacant jobs and unemployed persons from the perspective of efficiency. The assumption that returns yet another specification of the BC is that both v and u change approximately symmetrically. The BC is then a hyperbola in the (v, u) space:

$$vu = \alpha \quad (9)$$

As shown by Michaillat et al. (2022), for the US between 1950 and 2019, $\frac{d \ln(v)}{d \ln(u)} \approx -1$, satisfying the symmetry necessary to justify the hyperbolic interpretation. For Eurozone, a similar result, albeit on a much smaller dataset, was obtained by Lethienne (2024).

Sometimes, a multivariate approach is chosen to investigate the DMP framework, rather than focusing solely on v and u . This involves computing job-finding and job-separation rates from microdata, and then testing for correlation among the parameters. This is of use for looking at aggregate shocks to the economy, including recessionary periods, to test for cyclical properties of tightness and its determinants. The seminal study by Shimer (2005) found that job-finding rates for US were highly procyclical, while tightness, on the contrary, went against the economic trend. What is surprising is that the empirical volatility of tightness contradicts the assumptions in the original DMP framework, according to which productivity shocks, and hence the impact on vacancy creation, are absorbed by changes in wages. Hence, a methodology proposed by more recent econometric literature aims to examine the variables in the DMP model by modeling them as separate time series, susceptible to shocks, where structural and cyclical components are accounted for. Klinger et al. (2016), for instance, describe the evolution of matching efficiency, hiring rates, employment and vacancies for Germany using longitudinal data from 1980 to 2011. By modeling the unemployment time series as a combination of the time trends of multiple inputs to the matching function, each of which incorporates a stochastic trend and a cyclical component, the authors show how matching efficiency improved after the reforming of the German labour market in early 2000s.

The benefit of exploring the properties of the matching function is that it allows to better capture the contribution of particular variables to aggregate hiring. This may be useful to examine evolving trends in the interplay of these factors, or in the structure of the job market overall. However, there are serious drawbacks. First, positing a matching function to then isolate its parameters or the effects of separate variables requires a long-enough time frame, and a suitable-enough sample of observations. This is good for tracing out historical trends, but not as great for detecting immediate changes in the $v - u$. Certain parameters, such as the elasticity α , may be unstable and are themselves reliant on an a-priori specification of the matching function (Gottfries & Stadin, 2025). The latter necessitates a predefined formulation (some version of the Cobb-Douglas function is usually employed to this end), which must be adjusted over time, and is subject to additional discrepancies where cross-country or cross-region comparisons are made. Some studies examine *flows* into and out of labour instead of taking more readily available stock figures. Although no doubt useful,

this exercise requires extensive use of microdata, further complicating research (M. W. L. Elsby et al., 2009; Hall, 2005; Shimer, 2012). Another issue concerning the functional form of any matching function is that it may suffer from endogeneity bias. This renders OLS estimates inconsistent, while alternative measures are not always feasible due to a limited data set (Hobijn & Şahin, 2013).

A simple and robust approach that is closer in spirit to our work and is capable of capturing both shifts in the BC and changes in its slope involves regressing the level of unemployment on some estimate of vacancy data (Valletta, 2005) and serves as a good workhorse model for broader macroeconomic analyses:

$$u_t = \alpha + \beta_1 v_t + \beta_2 v_t^2 + \mathbf{Y}_t^T \boldsymbol{\tau} + \varepsilon_t \quad (10)$$

where $\mathbf{Y}_t \boldsymbol{\tau}$ represents the complete set of time effects in the sample, and the squared v_t term is added to capture the convexity of the curve. The advantage of this specification is that it effectively captures the shifts in the BC, which is done with the use of time-varying dummies, without simultaneously concerning itself with a decomposition of the variables. Of course, the timing of the changes in the values of the dummy have to be predicted first. However, when the focus is mainly to document significant changes in the $v - u$ relationship, a prediction based on visual inspection of the time series often proves to be sufficient for the purposes of research (Bonthuis et al., 2013; Valletta, 2005). The quadratic term may or not be included, depending on the fit. Same or similar specifications are often presented in log-form for practicality (Benati & Lubik, 2014).

One possible concern that deserves a separate mention is the correlation of residuals that often arises in models involving time-series. There are several distinct ways economic literature deals with this. The simplest and most obvious solution which we also adopt is to include lags of the dependent variable, u , in the regression (Bonthuis et al., 2013; Ebeke & Everaert, 2014). This approach makes sense, given that unemployment in the Eurozone and other OECD countries evolves according to a time trend (Solarin et al., 2025). Other methods involve cointegration of the time series (Bova et al., 2018) and GLS (Kiss et al., 2022).

Panel regressions using OLS and time-varying dummies were performed with good results for Germany (Börsch-Supan, 1991), the UK (Jackman & Roper, 1987; Wall & Zoega, 2002) and Italy (Di Giorgio & Giannini, 2012). Conclusions served well to capture immediate shifts, although characterising long-term trends proved more challenging (Börsch-Supan, 1991). In all cases, dummies were informative of any changes in $v - u$ relationship, and in some cases, long-term dynamics could be traced out more clearly, consistent with real events like labour market reforms (Bonthuis et al., 2016). There is a series of studies that involve fitting panel regressions, taking into account institutional variables reflecting policy interventions (Bouvet, 2012). Although general results differ across European countries, some trends appear to be common. One such result that is consistent with our findings is that vacancies are more responsive to aggregate shocks in the economy than unemployment, inducing a loop-like movement of the BC after a crisis (Bernstein et al., 2020). These findings have held well for the EU, according to empirical data (Consolo & Da Silva, 2019).

A spike in the relevancy of econometric studies involving BC occurred after the Great Financial Crisis (GFC) (2007-2011). With the help of broad specifications like those given above, large outward shifts of the BC for most OECD countries were well documented (Bova et al., 2018; Hobijn & Şahin, 2013). Factors like lower quit rate and poor performance of individual sectors of the economy, such as construction, were found to be especially insightful (Hobijn & Şahin, 2013).

The case of Covid is different, and the associated econometric studies have accounted for this in two ways. First, while the use of unemployment as a measure of labour supply is quite standard, vacancies are frequently substituted for labour shortages as a proxy of demand. Second, the baseline is augmented by the addition of regressors that reflect the particularities of the Covid crisis. Kiss et al (2022) include a skill mismatch and a sectoral dispersion indicator in the model. The former variable, defined as relative dispersion of employment across skill groups as outlined by the European statistical methodology (CEDEFOP, 2013), was found to be significant and contributing positively to unemployment for each level of vacancy rates.

The final conclusion of the paper was that an outward shift in the Euroarea's BC following the pandemic is temporary and does not point to a growing structural issue. Another experimental regressor, claims on insurance, has been suggested as a predictor of job posting by firms in the UK during 2020, when Covid struck (Cepparulo, 2024).

Quality econometric research for the post-Covid labour market is still scarce, perhaps due to its novelty. However, models testing the inverse relationship between vacancies and unemployment that incorporate fixed-effects and use either OLS, GLS, or both, have so far provided well-interpretable results for both the national and the EU-wide levels, and for periods of relative macroeconomic stability or during a major recession like the GFC (Börsch-Supan, 1991; Sell & Reinisch, 2013). We thus rely on a well-established framework to obtain a viable interpretation of recent labour market developments in Germany, UK, Italy and the Eurozone at large.

3 The Beveridge curve in Europe: a policy perspective

3.1 European labour markets before and after Covid

3.1.1 A general overview

As a starting point for the analysis, it is useful to look at historical data for the EU. An inspection of the relevant time series (Fig. 4) confirms the negative relationship between slack and job openings for the region (Leythienne, 2024). Since 2010, unemployment and vacancies have followed opposite trends: the former decreased while the latter rose. These trends temporarily reversed around 2020, a direct effect of the Covid epidemic. Interestingly, vacancies reacted more strongly than unemployment, which explains the loop-like movement of the curve during that period (Fig. 5). The movements of the BC and the changes in the tightness parameter for the region between 2007 and 2024 are illustrated in Fig. 5.



Figure 4: Unemployment vs. Job Vacancy Rate (JVR) in the Euro Area, 2007-2022. Source: Leythienne (2024)

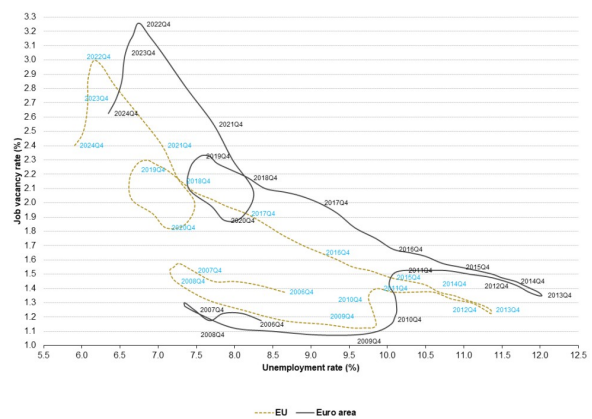


Figure 5: Beveridge Curve for the Euro Area, 2007-2024. Source: Eurostat (2024)

Fig. 5 plots the BC of the Eurozone (right curve) and the entire EU (left curve). The first substantial shift occurred in 2009–2013, when Europe was hit first by the Great Recession and then by the sovereign debt crisis. As a result, matching efficiency deteriorated, causing a large outward shift in the curve—a finding consistent with post-Covid scholarly literature (Barnichon & Figura, 2015). Between 2015–2019, the BC shifted inwards again, as job matching partly recovered (European Commission, 2024). During this period, labour supply outpaced the rise in demand (Weber & Adăscălișei, 2021). The onset of the pandemic in early

2020 corresponded to a steep fall in vacancy rates, with a much smaller impact on unemployment, as noted earlier. Why did jobs and employment not react equally strongly to the crisis?

One explanation relates to the vacancy rates themselves. The industries most affected by the shock, such as accommodation and food services, faced the most severe hiring difficulties, which drove down aggregate vacancies (European Commission, 2024). As reported by OECD (2021), the impact on these sectors extended beyond the initial three quarters most affected by lockdown, as many vacancies were never effectively reopened. However, the speed with which job postings and hiring fell is unlikely to be the only explanation for the discrepancy visible in Fig. 4. A closer look at 2020 data suggests that the response of unemployment was not only small relative to vacancies, but also relative to its usual reaction to a large macroeconomic shock.

The first and most important reason for the sluggish adjustment of unemployment during Covid is that, in most European countries, the change in labour activity occurred along the intensive rather than the extensive margin. In other words, the loss in labour was not due to workers losing their jobs, but rather jobs “losing” their workers. More precisely, labour retention schemes that became widespread in the wake of Covid allowed workers to work fewer hours without being laid off (OECD, 2020). This adjustment is not reflected in aggregate unemployment figures, and only becomes visible when examining the distribution of hours worked. Some workers saw their hours greatly reduced, while others stopped working entirely. The biggest “losers” in the latter category were concentrated in the same sectors where hiring was most affected (Fig. 6).

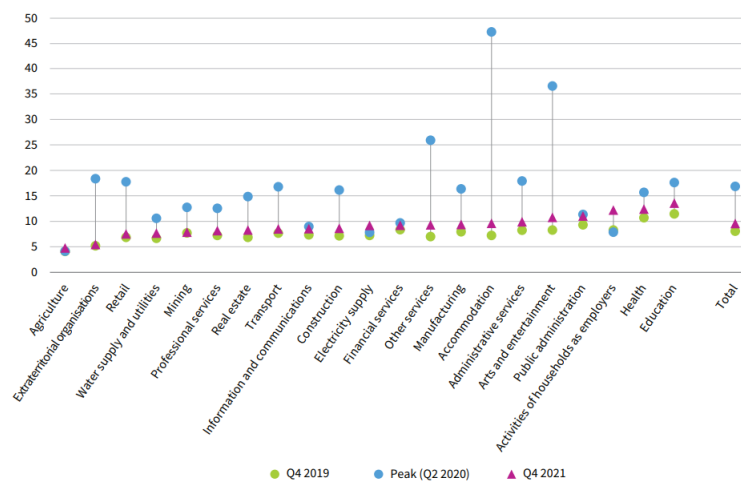


Figure 6: A sectoral breakdown of % of people employed but not working in 2019 Q4, 2020 Q2, 2021 Q4. Source: Eurofound (2021)

Following the initial shock, European governments implemented a variety of policies to cushion the economic impact of the pandemic on workers and employers. These went beyond standard unemployment insurance and included a wide range of wage and tax subsidies, as well as new tools not available before 2020. Thanks to these measures, much of the European workforce was able to return to the same workplace after lockdowns ended, as layoffs were substituted by reductions in employees’ workloads. At the same time, firms avoided a permanent loss of their workforce, while limiting payroll costs thanks to state compensation for lost labour, as well as the very nature of the adjustment (Von Wachter, 2020). Crucially, workers at risk of job loss were able to preserve some of their income while working less, which was particularly helpful for groups whose occupations were most exposed to the crisis. Insurance policies enacted across most European states in the first half of 2020 provided further relief, with funds mobilised through the EU’s Support to Mitigate Unemployment Risks in Emergency (SURE) temporary facility (Ando et al., 2022).

In general, the use of insurance and other benefits to prevent job losses was much more prevalent than during the GFC (OECD, 2020). Policy tools deployed by European governments as social safety nets can be divided into several categories (Ando et al., 2022):

1. Job retention schemes (JRS), allowing employees to retain their jobs during lockdowns. This was the most common way to mitigate negative effects on workers' earnings, implemented across most OECD nations, including all countries considered in this paper. JRS can be further divided into Short-Term Work Schemes (STW), which subsidised both workers and employers for unworked hours, and Wage Subsidies (WS), which targeted certain companies and were distributed as lump sums regardless of hours lost.
2. Hiring subsidies (HS), compensations paid to employers for new hires. These were distributed selectively, based on firm size or the type of worker recruited.
3. Wage-loss insurance, supporting employees forced to switch to lower-paying jobs.

Hence, the EU labour market adjusted predominantly through the intensive margin: hours worked per person declined more than total employment. At the same time, the distribution of reductions varied widely across workers, depending not only on sector but also on personal and professional characteristics (Fig. 7).

Sector	Q4 2020–Q4 2021			Q4 2020–Q4 2021		
	Female			Male		
	15–24	25–54	55–64	15–24	25–54	55–64
Accommodation and food service activities	0.8	3.4	3.3	-0.1	1.8	2.3
Administrative and support service activities	-0.3	0.1	-0.5	-0.9	0.1	0.4
Agriculture, forestry and fishing	1.5	0.8	2.0	0.9	2.0	2.6
Arts, entertainment and recreation	1.7	1.1	1.9	-0.8	-0.2	1.4
Construction	-0.8	-0.2	0.3	-0.5	0.6	0.5
Education	-0.3	-0.7	-1.2	-1.8	-0.7	-0.2
Electricity, gas, steam and air conditioning supply	-2.4	-0.2	-0.4		-0.7	0.3
Financial and insurance activities	-2.9	-0.4	-0.1	-1.2	-0.4	0.1
Human health and social work activities	-0.9	-0.5	-1.1	-0.4	-0.5	-0.6
Information and communications	-0.5	-0.2	-1.1	-2.0	-0.2	0.5
Manufacturing	-0.4	-0.4	0.0	-0.8	-0.2	-0.3
Other service activities	-1.5	-0.5	0.5	0.5	0.3	0.7
Professional, scientific and technical activities	-1.2	-0.1	0.1	-0.9	-0.1	2.0
Public administration and defence	-1.6	-0.5	-0.9	-1.4	-0.7	-0.3
Real estate	-3.3	0.0	0.9	3.0	1.0	-0.7
Transport and storage	-0.3	-0.1	0.0	0.5	0.8	0.9
Wholesale and retail trade	-0.4	-0.1	0.2	-0.5	0.1	-0.1

Figure 7: Year-on-year change in weekly work hours by industry, age and gender, 2020-2021. Source: Eurofound (2021)

A second reason behind the relatively muted response of unemployment is that changes in aggregate labour supply are not fully captured by the unemployment statistic. Inactivity, another key structural factor, rose after the crisis and, unlike the immediate declines in hiring and hours worked, adjusted more slowly. Transitions from unemployment into inactivity accelerated in the final two quarters of 2021, when 3.5 million unemployed individuals exited the labour force across the EU (Weber & Adăscăliței, 2021). This suggests that inactivity absorbed workers who might otherwise have been counted as unemployed, masking the true extent of labour market distress.

Cross-country disparities further illustrate this point. Italy, for instance, experienced larger and more persistent outflows from employment into inactivity in 2020–2021 than many of its European neighbours. This was due in part to the informal structure of Italy’s labour market, which left many discouraged workers exiting the labour force entirely (OECD, 2021). The rise in “discouraged” workers, who faced long unemployment spells and were more likely to withdraw from the labour pool, was reflected in reduced job search intensity throughout 2020. Quantitatively, this appeared in a decline in online job search activity, as measured by the frequency of relevant search prompts (OECD, 2021). As a result, labour market matching remained subdued not only at the height of lockdowns (2020 Q2) but also in later periods.

Despite the severity of the shock, the loop-like movement of the BC between 2019 Q4 and 2020 Q4 indicates that the European market began recovering relatively quickly. The medium- and long-term response of the labour market is also noteworthy and can be seen in subsequent movements of the BC (Fig. 5). Already in 2021, the trend of rising market tightness that began in 2015 resumed. As economic activity picked up, the $v - u$ ratio climbed steadily, peaking in Q4 2022. For much of the post-crisis period, European labour markets were tighter than before 2020 (OECD, 2025b).

Based on these developments, several conclusions can be drawn about Covid’s overall impact on the EU labour market. First, the immediate consequence was a substantial decrease in job postings, with only a modest increase in unemployment. This distinguishes Covid from previous crises of similar magnitude, most notably the GFC, when the labour market adjusted primarily through labour supply rather than demand. Second, the recovery was robust but uneven across countries, as demographic and structural factors pre-dating Covid influenced recovery paths. Italy, for instance, was more exposed to shocks in hospitality and tourism than Northern countries such as the UK and Germany. These disparities also affected different age groups: young workers, many employed part-time in these sectors, suffered more than the mature workforce (ILO, 2023; OECD, 2023b). Third, as the “tail” of the curve between Q4 2023 and Q4 2024 shows, the labour market eventually cooled from its peak, although unemployment settled at a lower level than before 2020. Once again, changes in tightness were driven more by labour demand than by supply, as hiring fell. Finally, Covid may have exacerbated pre-existing skill gaps and contributed to the post-pandemic rise in labour shortages. However, this conclusion remains debated, and its impact on the $v - u$ relationship is not yet clear (Kiss et al., 2022; World Bank, 2022).

3.1.2 Germany

In Germany, the impact of Covid on its labour market was most pronounced in the second quarter of 2020, when both unemployment and vacancy rates reacted dramatically (Fig. 9). The percentage changes with respect to the first quarter stood at -31% and +12% for JVR and U, respectively (Eurostat, own calculations). Prior to the crisis, the Germany was steadily moving toward a tighter market, with both series gradually moving in opposite directions (Fig. 8).

What is also evident from the figures, is that the labour dynamics in Germany have been overall resilient, despite a brief break induced by Covid. The recovery was strong, and between 2019 Q4 and 2020 Q3, unemployment, despite exhibiting a large relative increase, did not change much in absolute terms. The change over this period was less than 1%, which implies that the storm was weathered well. There is a consensus in the literature that the policies enacted by the German government as a response served as a catalyst to this recovery, in addition to preventing other negative effects such as widening inequality (Braband et al., 2022).

In particular, *Kurzarbeit*, a STW scheme which was the core of the country’s response to the crisis, acted as a cushion for the German labour force throughout the recessionary period. *Kurzarbeit* was introduced in Germany during the Great Recession, but the access to the scheme was streamlined in the wake of Covid. As a typical STW scheme, it allowed workers to work less while being compensated by the government for the resulting loss of income. During the first two quarters of 2020, which were most marked by lockdowns,

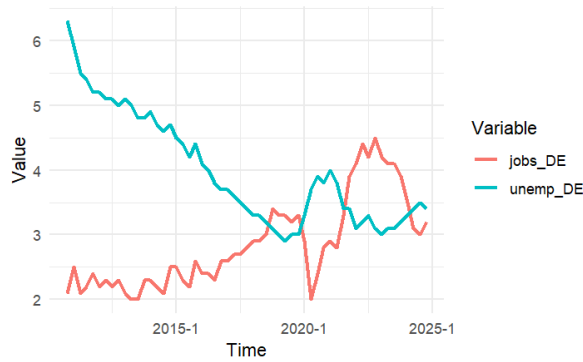


Figure 8: Unemployment and vacancy series for Germany, 2010-2024(quarterly). Source: Eurostat, own elaboration

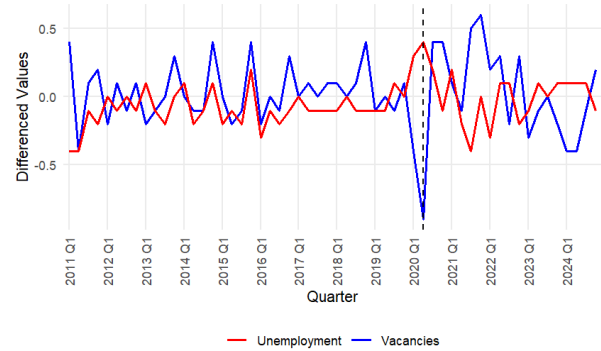


Figure 9: Differenced series. Source: Eurostat, own elaboration

nearly a fifth of the employed population was covered by the scheme. The share of low-income workers in the total number of beneficiaries was higher than that of other income groups, which helped to mitigate the negative effects of the crisis on the most vulnerable segments of the population (Braband et al., 2022).

Still, Covid impacted different segments of the market, both on the demand and supply side, unevenly. As everywhere in the EU, low-skill and low-wage workers were the most affected. Despite the generous coverage of Kurzarbeit, over 20% of workers who had been employed in 2019 Q4 and fell under the aforementioned category found themselves jobless by mid 2021 (Braband et al., 2022).

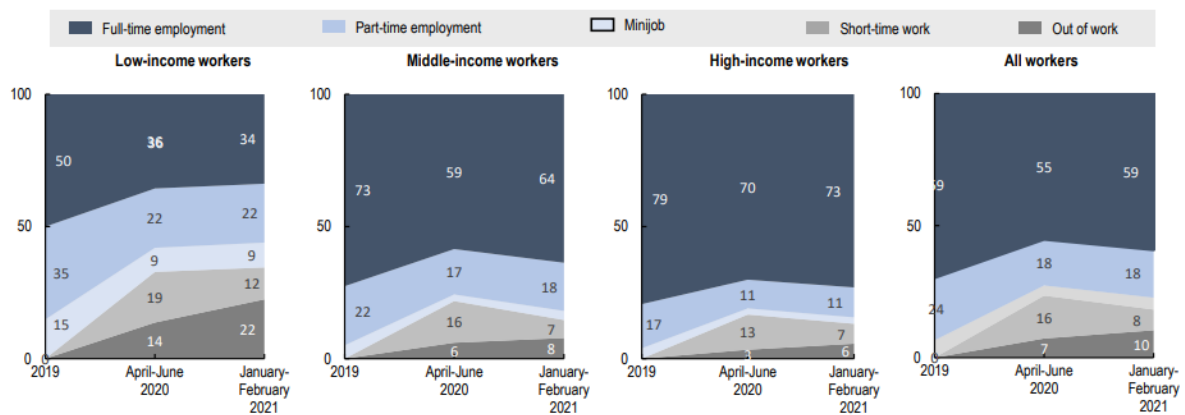


Figure 10: Share of German workers in part time and full time employment by income group, 2019-2021. Source: Braband et al. (2022)

The reallocation of workers between part-time and full-time work was also unequal for different segments of the German workforce, with low-income employees staying in part-time jobs longer than their higher-income counterparts, even in the post-peak phases of the crisis (Fig. 10). This suggests that the group was exposed to a higher degree of job insecurity after Covid, which may have affected the matching process beyond 2020 and 2021.

Another interesting observation that concerns the impact of Covid on the composition of Germany's labour force and is implied in Fig. 10 is the increase in the proportion of people that fell out of work entirely. By the end of 2021, the share of population in inactivity doubled from 0.5% to 1% compared to the pre-Covid period. According to subsequent estimations, this was not a result of Germany's preexisting demographic

trends, but rather a direct consequence of the crisis (IMF, 2023). However, in terms of inactivity rates observable 2-3 years after the onset, the German labour market actually fared better than before Covid. By 2023, the rate of participation for the country not only returned to the pre-Covid levels, but exceeded them - a sign of *Kurzarbeit*'s and similar policies' success, as well as the general resilience of the German economy (OECD, 2023b).

Still, Covid posed new challenges for the outlook of German labour dynamics in the long run, while adding to those existing before it. The issue that has been mentioned especially frequently in the academic literature on the legacy of Covid is the skill mismatch - a term referring to the widening gap between the skill of the unemployed and those sought by employers. According to OECD, in 2024, about a third of the occupied German population were employed in jobs that did not match their education level (Organisation for Economic Co-operation and Development, 2024). Despite both the employment and activity levels having risen overall by 2024, reflecting a tighter market, the share of firms reporting labour shortages approached 40% and was the highest among all OECD countries (OECD, 2025a). This was not a new phenomenon for Germany, as it was partly caused by secular trends, such as the aging of the country's workforce and the lack of high-skilled professionals in the demanded occupations. Covid added to these developments by reshuffling the workers both across industries and jobs within the same industry. Labour shortages two years after the crisis had struck were concentrated in several sectors: healthcare, civil engineering, construction, hospitality and transport (OECD, 2025a). Another key factor, related to the skill-based discrepancies between worker groups and the labour shortages, was the reduction of average working hours per individual. Contrary to the other discussed indicators, the intensive margin is one trend that was affected by Covid not only in a manner that was quantitatively significant, but also lasting. The share of workers in part-time employment increased by 10% from 2019 to 2023 (OECD, 2025a).

Overall, Germany's market recovered well from Covid, and its pre-crisis movement toward higher employment resumed by 2021. Since its outward shift during the crisis, the BC has largely moved along and even past its pre-Covid tightening trajectory. Peak tightness was reached between 2021 and 2022, but the inward movement of the curve did not stop until 2024. This is consistent with what was observed at the EU-wide level, as described in the previous section (Fig. 11).

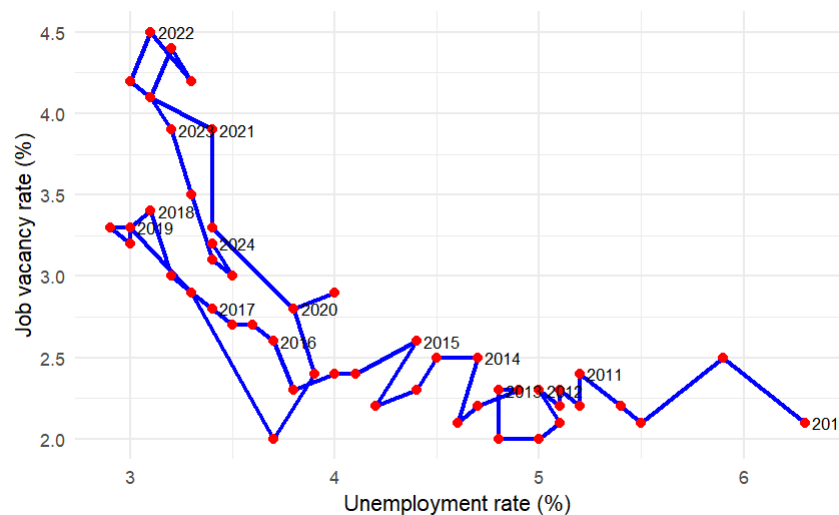


Figure 11: BC for Germany between 2010 and 2024, quarterly. Highlighted points denote Q4 of each year in the sample. Source: Eurostat, own elaboration

However, microdata suggests that there has also been an opposite effect on labour supply, which manifested along the intensive margin, as average working hours have declined since Covid. Partly this may

be attributable to technological and productive changes not tied to the pandemic. Other trends, though, are likely part of the crisis' lasting legacy. First, remote working, used by firms to sidestep lockdown-induced strains on their employees, likely made working arrangements more flexible also in the mid- and long-term, especially for the highly skilled segment of the labour force (ILO, 2023). Second, Covid has brought about a reallocation of workforce to part-time jobs, which typically coincides with lower skill and education requirements. The latter may prove to be the shadow side of STWs in Germany, as shortages concentrated in high-skilled sectors are now less likely to be resolved. Coupled with an ageing population and a changing economic landscape, this may lead to persistent structural mismatch in Germany's labour market in the future.

3.1.3 UK

Same as in the German case, UK had steadily moved towards a tighter labour market prior to the onset of the crisis. The timing of the strongest response in both labour demand and supply, as proxied by the vacancy rates and unemployment, was also similar, except that the latter reacted one quarter later - in Q3 2020. The figures for the timing and magnitude of these developments are given below.

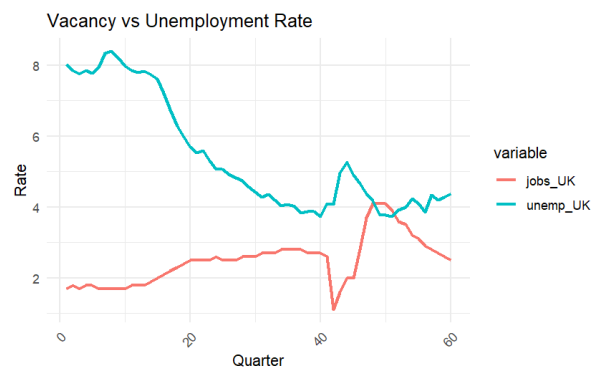


Figure 12: Unemployment and vacancy series for UK, 2010-2024(quarterly). Source: ONS, own elaboration

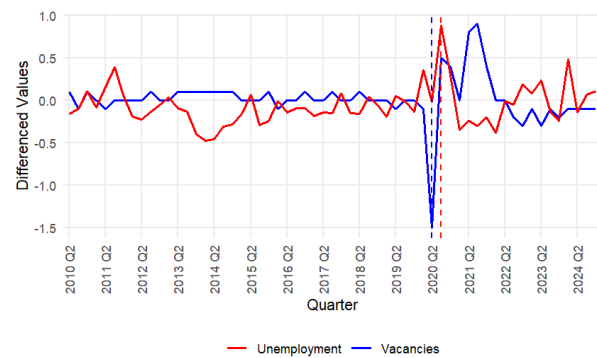


Figure 13: Differenced series. Source: ONS, own elaboration

The labour demand reached its trough in the second quarter of 2020, as the quarter-on-quarter decline was 58%. Slack spiked in Q3 and Q4 2020, reaching a local peak of 5.26% (ONS, own calculations).

A good summary of the UK labour market developments during and shortly after the onset of Covid is given by Cepparulo (2024). He identifies 3 main stages in the dynamics of UK labour in the post-pandemic period:

1. The initial shock with a rapid decline in demand, accompanied by lockdowns (March-May 2020). This was the phase during which STWs and similar policy measures were implemented by the British government. The two primary measures were the introduction of the Coronavirus Job Retention Scheme (CJRS) and the Self-Employment Income Support Scheme (SEISS). The former scheme, in particular, aimed at shielding the self-employed, which is something that was largely absent from the German policy framework (OECD, 2023a).
2. The 'mismatch' phase (Q2 2020 - Q1 2021). As the shock was unequally distributed among industries, there were notable differences in the sectoral distributions of employment during this period. Unsurprisingly, the most hard-hit sectors were hospitality, entertainment and retail. However, research

focused on this stage suggests that the impact of the pandemic on mismatch unemployment was temporary (Pizzinelli & Shibata, 2023). The aggregate effect on the BC in the mismatch phase was that of a large outward shift (Cepparulo, 2024).

3. The inactivity phase, which was evident throughout 2021-2022 and featured large net flows out of active workforce. Cepparulo ties such developments with a broader demographic issue faced by the UK, as older generations resigned in large numbers. This distinguishes the UK labour market from other OECD nations, including Italy and Germany, where outflows from active workforce were neither as large, nor as persistent in Covid's aftermath. This is likely to have profound implications for the future of the country's employment, as policies focused on the reskilling and upskilling of contemporary workforce call for subsequent augmentation (BEISC, 2023). Despite a partial recovery in aggregate activity after 2023, at the start of 2025 the overall figure of 21.4% was still a full percentage point higher than in Q4 2019.

As in other countries, much of the adjustment in labour supply during Covid occurred along the intensive margin (Fig. 13).

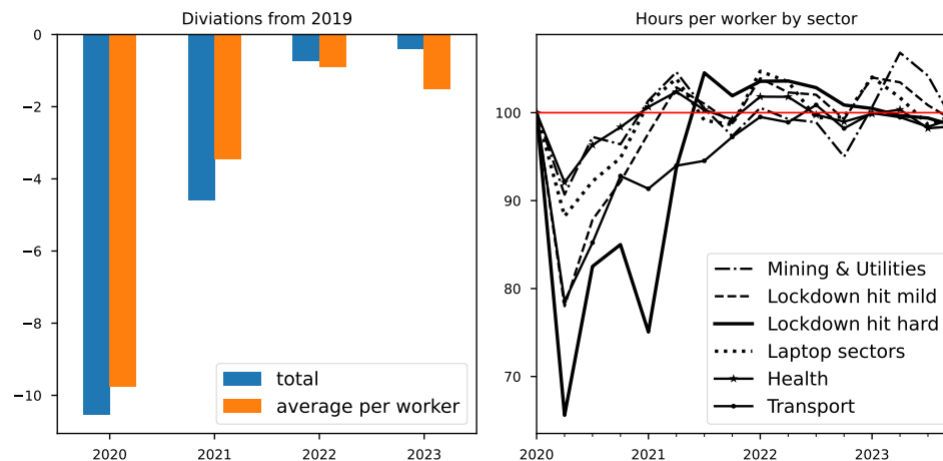


Figure 14: The decrease in average total and per-worker annual hours worked in UK, 2020-2023, aggregate and by sector. Source: Cepparulo (2024)

What is of note is that the working hours remained subdued and declined even further in 2023, way past the lockdowns' peak three years prior. A possible explanation, given in Cepparulo (2024), is that this 'silent quitting' was caused by premature voluntary resignations of older workforce. On average, both UK's men and women retire around 3 years before reaching the normal retirement age, leaving gaps that are likely to cause more shortages and tightness. This is further supported by the fact that the age of an average British employee is above OECD's mean working age (OECD, 2025b). To mitigate these dynamics and stimulate higher levels of activity, in November 2024 the British government presented a new strategy, aiming to bring the employment rate to 80% (DWP, 2010).

The segments of the UK's labour force that suffered the most during the active stages of Covid consisted primarily of those in low-paid occupations, as well as the youth. While the damage was contained thanks to the timely job retention schemes and employment support, these groups were still subjected to higher job and income losses. However, the recovery among the aforementioned categories of workforce was relatively robust, and by Q1 2022 the unemployment figures for both low-paid and young workers returned to their 2019 levels (Foley et al., 2020).

In the BC framework (Fig. 14), the period from 2020 to 2024 was marked by a large but brief outward shift, which reversed in Q1 2021. During 2021, the curve steepened, driven by a surge in job demand. In 2022, the BC moved inwards, as unemployment contracted sharply, which caused the market to reach its peak tightness. From 2023 to 2024, the tightness indicator gradually decreased, approaching its pre-Covid value.

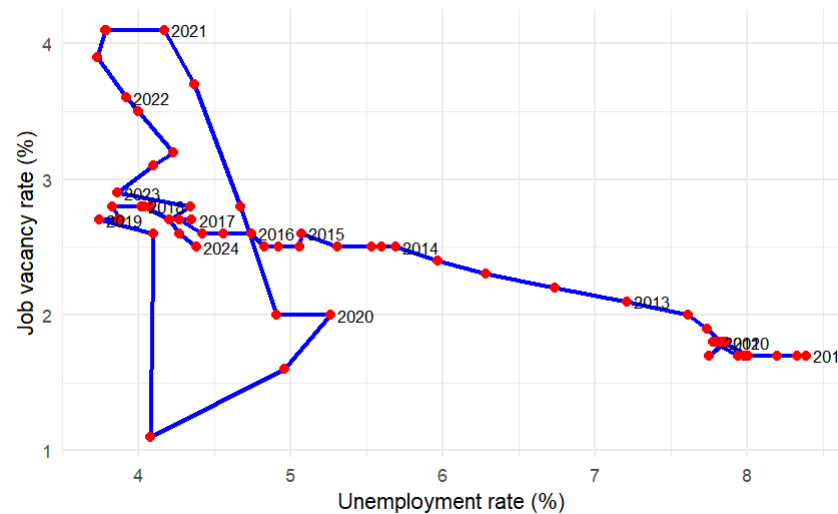


Figure 15: BC for UK between 2010 and 2024, quarterly. Highlighted points denote Q4 of each year in the sample. Source: ONS, own elaboration

The last phase of decreasing tightness and increasing unemployment reflects a decline in employer confidence, and a rise in the costs of recruitment which may have pushed firms to scale back on previous hiring efforts (CIPD, 2025). The long-term outlook for the UK's labour market will depend largely on how the nation's economy handles the issue of a rapidly ageing workforce, which may exacerbate the existing shortages of labour and push the tightness parameter up again (DWP, 2010).

3.1.4 Italy

The data for Italy, the last country in our sample, indicates that vacancy and unemployment rates reacted most prominently in 2020 Q1 and 2020 Q3, respectively. In percentage terms, these responses were of -62% and +33%, exceeding those of Germany and the UK (Fig. 16, 17).

During 2020, Italy suffered qualitative effects that largely resembled those that could be observed in the UK and Germany. However, the Italian economy stood to lose even more from the Covid-induced restrictions due to its reliance on tourism, hospitality and entertainment. As everywhere across the OECD, the recovery in aggregate labour demand, which began in 2021, was also driven mainly by vacancy creation in those sectors. Another feature that made the Italian labour market more vulnerable to the shock was the large proportion of its temporary and self-employed workers in the total employment pool. Combined with the fact that the Italian system of social security is structured in a way to prioritize the protection of workers on permanent rather than temporary contracts, this led to additional risks for the most vulnerable categories of population (Carta & De Philippis, 2021). The primary responses of the Italian government were the implementation of insurance in the form of STWs (as in other countries), and a lump-sum pecuniary bonus, which was distributed among the Italian households during the spring of 2020. The latter was specifically designed to address the negative impact of Covid on the incomes of the self-employed. Despite their progressive nature, these policies were able to offset income losses and the growing inequality between different socioeconomic

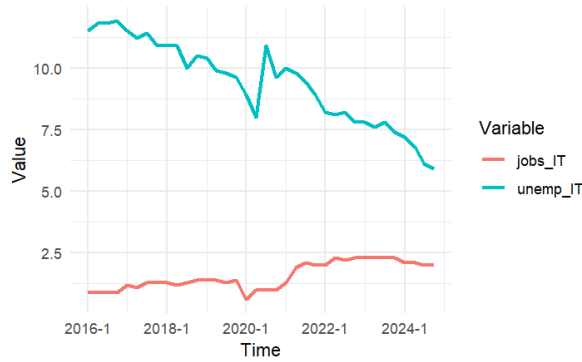


Figure 16: Unemployment and vacancy series for Italy, 2016-2024 (quarters). Source: ISTAT, own elaboration

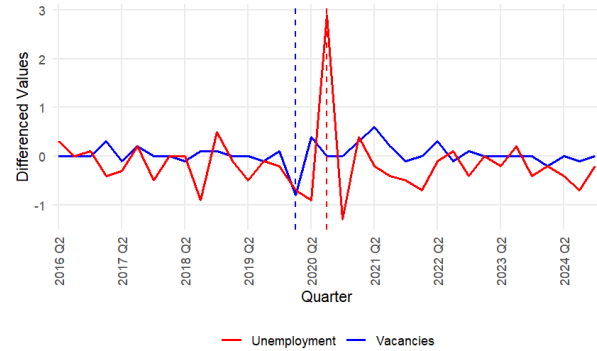


Figure 17: Differenced series. Source: ISTAT, own elaboration

groups only partially for the first (and the most acute) 2 quarters of 2020 (Carta & De Philippis, 2021). However, studies find that overall between 2020 and 2021, government subsidies contributed to a slight but measurable reduction in the absolute and relative inequality indicators for Italy from their 2019 levels (Clark et al., 2021). This suggests a potential role of such policies in mitigating the adverse effects of future crises on economic inequality, in addition to their usual tasks of preserving socioeconomic stability and stimulating the aggregate demand.

The contribution of the intensive margin to the overall job loss in Italy between 2020 and 2021 was among the largest among all OECD members, exceeding the figures for the UK and Germany. According to the OECD, this contribution, measured as the reduction in average hours worked per week multiplied by total employment, totaled 90% for the Italian economy in 2020. (OECD, 2020).

Another key challenge posed by the crisis was its effect on the unemployment rate of the Italian youth. As everywhere in Europe, young workers were more susceptible to losing their jobs due to the lockdowns and a declining labour demand. However, the Italian case is remarkable for the historically high share of its youth classifying as 'NEET' (Not in Education, Employment, or Training). Even before Covid, this figure exceeded 20% (OECD, 2020), and the pandemic caused an additional spike in the figure. Fiaschi et al. (2022) find that the flows into NEET for Italian residents aged 15-34 peaked in 2020 Q2, when the percentage of people in NEET for this group surpassed 25%. In general, the topic of NEET for Italy is well-known and covered extensively by the pre- and post-Covid literature (Alfieri et al., 2015; Contini et al., 2019; Quintano et al., 2018). The implication for the present analysis is that the country's labour dynamics after Covid were largely affected by factors that require a decomposition of aggregate unemployment into its age- and sector-specific constituents. A careful examination of these decomposed statistics is also helpful to understand why aggregate unemployment in Italy is less informative of the BC-related developments than is the case for the UK and Germany. However, the aggregate measures provide sufficient insight into the general trends in the country's labour market, enabling a characterisation of its dynamics within the BC framework in a manner similar to that for the other countries.

The Beveridge curve for the period between 2016 and 2024 is not as clear as that for the UK and Germany, and its slope appears to be much flatter. This implies a weaker $v - u$ relationship for the country compared to its European counterparts, which is consistent with the findings for earlier periods (Di Giorgio & Giannini, 2012). In spite of this, there is a visible outward shift of the curve in Q3-Q4 2020, which is similar but more drastic than in the UK and Germany (Fig. 15). After a rebound in job demand, which began in 2021, the curve moved inwards, as the labour market tightened. This also aligns with the observed trend in the EU, the UK, and Germany around the same time.



Figure 18: BC for Italy between 2016 and 2024, quarterly. Highlighted points denote Q4 of each year in the sample. Source: ISTAT, own elaboration

A look at labour dynamics from 2021 to 2023 reveals several interesting patterns. For 2021, the Italian market experienced a significant increase in job reallocation, a measure of 'churning' in the market that is known to rise during major economic shocks (Bleakley & Fuhrer, 1997). Ciotti et al. (2023) show that in 2021, job-to-job transitions increased for all sectors of the Italian economy relative to 2019, especially for manufacturing, construction and services. The authors call this 'the great reallocation', a phenomenon that had not occurred in Italy, at least on such a scale, prior to the Covid shock. In 2022, job churning remained remarkable, although now driven mainly by dismissals rather than voluntary separations. Ciotti et al. (2023) tie these developments mainly to the start of the war in Ukraine and the subsequent energy crisis, to which the Italian industry responded with large-scale layoffs, but Covid may have played an additional role.

Despite a recovery of unemployment and a slight rise in activity rates beyond the pre-crisis level by 2025, Italy's labour market has faced serious challenges in the wake of the pandemic. Demographics is one issue, as working-age population in the country is projected to decline by over 30% in the next 40 years (OECD, 2025c). Another issue is the share of population in NEET, which remains substantially higher than the OECD's average. The figures are heterogeneous across different Italian regions, reflecting the productivity division between South and North, which further complicates aggregate analysis. Fiaschi et al. (2022) highlight the disparities in post-Covid labour dynamics between South and North-Center Italy by examining the share of population in various employment states in both geographical areas. Their findings show that female activity rates in North and Central Italy dropped more compared to the southern regions, due to higher participation rates for Italian women in the North before 2020. Transition rates across employment states were also unequal, as the number of people aged 20-29 who had been in education prior to the crisis was more affected in the South, where the labour market is weaker. The figure for total employment for South Italy was below 50% in 2020, when the outbreak first occurred, while in the North and Central Italy it stood at 65%, which again highlights these regional differences (Destefanis et al., 2021). There is also a large gender gap in participation in the South, which trended around 30% between 2007 and 2020 and exceeded the equivalent statistic for the North and Center by 10% (Destefanis et al., 2021). As a consequence, the impact of the shock in terms of employment was more pronounced in the North and Center, although over the long term, the South may have been affected in terms of subdued activity and further disincentives to work. Further research is needed to quantify the effects of Covid on the labour market in the Italian South, but the immediate implications of such geographical division for the aggregate Beveridge curve framework

are clear.

Labour shortages have also increased since 2020, as a third of the Italian current workforce has been described by the OECD as underskilled, and thus susceptible to being replaced by AI. While the medium-term impact of Covid on the tightness in the country's labour market appears to be modest, as unemployment is expected to remain stable at around 7% in 2026, and the vacancy rate has trended around 2% since 2021, many of the aforementioned structural challenges will likely affect Italy's employment dynamics in the long run (OECD, 2025c).

3.2 Lessons and implications

The Covid shock had a profound but relatively brief impact on the European labour market, according to traditional metrics of vacancies and unemployment. In the first three quarters, hiring decreased disproportionately across sectors in all EU countries, with hospitality and tourism being hit the hardest. It also affected the quality of jobs and job security, especially for employees in low-wage occupations and for young people on temporary contracts. According to a survey by Eurofound (2020), by April 2020, 8% of all employed individuals in the region had lost their jobs, and nearly half (49%) had seen their working hours reduced. This latter effect was particularly pronounced in Southern economies, including Italy.

Following the turbulent year of 2020, which saw the sharpest drops in hiring and reductions in employment along both the intensive and extensive margins, all indicators for the three countries began to recover. The recovery was driven mainly by job creation, concentrated in the same sectors that had suffered the most in the immediate aftermath of the crisis. Unemployment, although partially offset by worker support schemes, remained above 2019 levels for most European countries throughout 2021 (OECD, 2021).

Policy responses focused primarily on grants and subsidies for workers and employers, as well as the aforementioned STWs, although additional measures such as partial VAT reductions for small enterprises were also implemented. These programs were similar across the three countries, with Germany being the most generous in terms of total funds allocated (International Monetary Fund, 2020). They were instrumental in preventing mass layoffs and a large-scale social crisis, helping to preserve around 20 million jobs in the OECD region (Hijzen & Thévenot, 2022). Studies also find a positive effect of subsidies targeted at low-income groups in reducing both absolute and relative inequality in Italy (Clark et al., 2021). Where job-retention schemes already existed before Covid (such as *Kurzarbeit* in Germany), their uptake was easier and more efficient, and recovery more robust, suggesting the value of such facilities in the future (Hijzen & Thévenot, 2022).

However, STW and similar schemes also created challenges. Upskilling and reskilling—crucial for addressing long-term shortages in European labour markets—were hindered by reduced work hours and lower job search intensity (Hijzen & Thévenot, 2022). Barrero et al. (2020) explore the effect of the so-called "job reallocation" in the US, characterised by movements between different jobs during Covid. In Europe, disparities in sectoral employment widened in 2021, leading to skill and employment mismatches across all three countries (Braband et al., 2022; Casarico & Lattanzio, 2022; Cepparulo, 2024). For the UK and Germany, these mismatches took the form of worker flows from more affected to less affected sectors, which reversed by the end of 2020, while Italy faced the additional challenge of the less developed South falling further behind the more developed North. Across all countries, Covid also underscored the digital divide, where reskilling efforts were hampered by insufficient digital competencies among the working-age population (OECD, 2021).

The indicator of labour market tightness, which reflects the evolution of the Beveridge curve before and after the crisis, adjusted sharply downward in the first post-Covid quarters, before trending upwards and peaking at the end of 2022. This was largely due to the recovery in job demand, with retail, hospitality, and service industries rebounding as discussed in the previous section. Between 2023 and 2024, labour markets remained tighter than before Covid, while inactivity fell below 2019 levels in Germany and the UK. The UK

also experienced a wave of voluntary resignations from 2021, driven mainly by workforce demographics rather than Covid, which further contributed to shortages and tightness (Cepparulo, 2024). Aggregate data suggest that by 2023, European economies had largely recovered, although some longer-lasting effects persisted, such as the higher share of short-term work in Germany. Academic literature remains divided on the extent to which Covid explains post-2022 labour developments, though some authors argue that the shock to the Beveridge curve was mostly temporary (Kiss et al., 2022).

Long-term changes in the structure of European labour and the nature of work, which are common to all three countries and to which Covid may have contributed, can be summarized as follows:

1. **Changing working habits.** Employees have increasingly moved toward flexible arrangements that allow remote work. Although teleworking peaked in 2022, it left a lasting legacy in the form of hybrid work, combining office hours with work from home (Causa et al., 2022). Covid was not the sole driver of this transition, but it accelerated its adoption from 2020 onwards. The early success of remote work was welcomed by both employers and employees. Barrero et al. (2021) find that workers in the US valued a hybrid arrangement as much as an 8% salary increase. Across the OECD, the share of teleworking doubled compared with 2019 levels in the three years following the crisis (OECD, 2023c). Evidence from the demand side supports this trend: online job postings reveal a higher share of vacancies offering remote options compared with 2019 (Cœuré, 2020). While technological change and digitalization have also promoted hybrid schemes (OECD, 2023c), the role of Covid is undeniable.
2. **Reallocation between industries.** Workers increasingly moved away from contact-intensive sectors. Studies conducted after 2022 suggest a permanent shift of the workforce away from hotel and restaurant businesses (Aiyar, 2023; Causa et al., 2022). Worker preferences also played a role, with many switching jobs or industries due to poor working conditions and greater concern for work–life balance in the wake of the pandemic (Vyas, 2022). This twofold shock particularly affected retail, tourism, and manufacturing: first during the acute phase of 2020, when lockdowns triggered closures and layoffs, and again in early 2022, when worker preferences limited recovery despite the rebound in vacancies.
3. **Demographic decline and ageing.** The ageing workforce of OECD countries threatens productivity and is expected to tighten labour markets in the long run. The retirement of older workers coincided with the post-Covid recovery, accelerating this trend. Higher mortality rates among older populations due to Covid may have also contributed, though precise estimates are not available (Trias-Llimós, 2022). Rising shortages and tighter markets in 2023–2024 can therefore be linked largely to demographic dynamics (OECD, 2025b). The broader “great ageing” phenomenon in OECD countries is also likely to have profound consequences for future growth, reducing productivity, limiting labour mobility, and worsening job matching (OECD, 2025b).

Thus, although the short-term impact on the Beveridge curve and the overall state of labour in the EU and the three countries considered in this work was successfully mitigated by job-retention schemes, persistent changes in work arrangements and the composition of labour supply remain a relevant topic for future research, and a potential challenge for European policymakers.

4 Econometric analysis of the shifts in Beveridge curves

4.1 Methodology

To test for statistical significance of the observed shifts, a dynamic linear model is applied to a basic Beveridge curve specification:

$$U_t = \beta_0 + \sum_{i=1}^p \beta_{i,1} U_{t-i} + \sum_{j=0}^q \beta_{j,2} L S_{t-j} + \beta_3 D_t + \epsilon_t \quad (11)$$

The method described in Bonthuis et al. (2016) is followed, with the difference that the job vacancy rate is used instead of alternative metrics — a choice explained in the next section. The dummy variable D_t takes the value of 1 following the Covid crisis and 0 otherwise. The periods affected by the crisis are identified by examining job demand, which reacted more than slack for all three countries. As in Bonthuis et al. (2016), break points are determined separately for each country, although the job vacancy rate (JVR) is used instead of GDP to improve accuracy.

Logs are not employed, as they tend to perform better only for larger samples by smoothing temporary fluctuations. This is undesirable for capturing shifts, particularly in the context of Covid, whose effect on the $v - u$ relationship was strong but brief. Italy, in particular, is a country where modeling the curve over a period of 10 or more years was previously found to be problematic (Di Giorgio & Giannini, 2012).

For lag length selection, both AIC and BIC criteria are consulted, although smaller samples favor the former (Liew, 2004). For all three countries, this results in a suggested lag length of no more than 2 for both variables. Likelihood-based estimation should be treated with caution in small samples, and it is generally advisable to restrict the number of lags to avoid overfitting (Hamilton, 2020). Introducing lags beyond 1 can also affect regression estimates due to collinearity among predictors. However, this does not qualitatively affect the efficacy of the model. Unemployment persistency in OECD countries is well documented, and including lags is consistent with both economic and statistical reasoning. First lags of unemployment are statistically significant at the 1% level for all three countries, and their inclusion greatly improves the overall accuracy of the regressions. Lags of JVR, in contrast, contribute little to regression accuracy and are less consistent with economic theory. During Covid, vacancies reacted more quickly than unemployment, which further supports omitting additional lags of JVR. Pissarides (2000) notes that vacancies adjust more quickly than unemployment during shocks, as firms can open or close positions at a relatively low cost. Unemployment exhibits a high degree of hysteresis, or persistency (Contini et al., 2019; Pissarides, 2000). Therefore, additional lags on JVR are not included in the baseline model, except for Germany, where the first lag is retained.

Regression analysis is conducted using OLS, which remains unbiased in smaller samples and performs adequately with dummy variables. Visual inspection of time series and their differences indicates that they share the same trend and are cointegrated. This supports the viability of the method and its suitability for explaining shifts as well as the historical negative relationship between unemployment and vacancies.

4.2 Data

4.2.1 Choice of variables

Labour supply

Unemployment is the broadest measure of labour availability and has been used extensively in econometric studies of the Beveridge curve (Bonthuis et al., 2016; Jackman & Roper, 1987; Kiss et al., 2022; Valletta,

2005) as well as in policymaking. In the DMP model, equilibrium unemployment is expressed as a function of inflows and outflows into employment, which depend on exogenous processes (Poisson rates of job exits and job matches) as well as labour demand (vacancies). The levels of unemployment themselves also influence these relationships, as the number of hires increases at higher unemployment for two reasons: (i) given a stable matching rate (Poisson parameter), the absolute mean number of hires grows; (ii) at higher unemployment, reservation wages decline, reducing hiring costs for firms at each level of job openings. The primary purpose of unemployment is to reflect the availability of workers in the economy, which affects the degree of labour market frictions (M. W. Elsby et al., 2015).

The main consideration is whether unemployment accurately proxies labour availability. According to ILO (2023), a person is unemployed if all three of the following criteria are met:

1. Without work: the person is not engaged in paid employment or self-employment during the reference period.
2. Currently available for work: the person is able to start work immediately or within a short period.
3. Actively seeking work: the person has taken specific steps to find a job, such as applying to employers, registering at employment agencies, or consulting job advertisements.

Germany and Italy follow the Eurostat methodology, which essentially aligns with ILO guidelines. A person is considered 'currently available' if ready to work within two weeks, and 'actively seeking' if work was sought during at least the previous four weeks (Eurostat, 2025). The UK adopts a similar methodology as set by the Office for National Statistics (ONS) (ONS, 2025a).

Survey design, timing, and seasonal adjustment may introduce differences. All three countries implement the European Labour Force Survey (EU-LFS), but variations in sampling and panel design exist. Italy uses stratified multi-stage sampling covering households (ISTAT, 2025), whereas the UK uses a rotating panel with quarterly interviews (ONS, 2025b). Such differences may affect the measurement of short-term unemployment spells.

Survey timing may lead to state transitions between being asked and responding, as described in Shimer (2005). Transitions among employment, unemployment, and inactivity must be considered to account fully for labour market flows and resulting tightness. Although these issues may slightly influence short-term unemployment data, they do not compromise the main objectives.

Absence of labour participation data may marginally affect aggregate labour supply estimates. Shifts in unemployment in 2020 were apparent from the data. Inactivity rates between 2021–2023 were only slightly affected (OECD, 2025b). For developed economies outside Europe, transitions from unemployment to inactivity had a more pronounced impact. Assumptions that hires occur predominantly from the unemployed pool are confirmed by multiple studies (Blanchard & Diamond, 1989; M. W. Elsby et al., 2015). Therefore, unemployment, while imperfect, remains the most representative measure for labour supply.

The measure does not account for intensive employment margins. For countries with retention schemes (Germany, Italy), this may introduce sector-specific bias. Alternative methods using hours worked are not applied to avoid complexity and due to lack of precedent. However, these limitations are modest relative to the study's primary objectives.

Labour demand

The official job vacancy rates are used to ensure the fullest possible representation and to avoid gaps in the dataset. According to Eurostat (European Commission, Eurostat, 2023), a job vacancy is defined as a paid post that is newly created, unoccupied, or about to become vacant:

1. for which the employer is taking active steps and is prepared to take further steps to find a suitable candidate from outside the enterprise concerned; and

2. which the employer intends to fill either immediately or within a specific period of time.

The official definition measures vacancy rates as the proportion of open positions to the total jobs in the economy:

$$\text{JVR} = \frac{\text{Number of Job Vacancies}}{\text{Number of Occupied Posts} + \text{Number of Job Vacancies}} \times 100$$

There are, however, slight differences in how data are collected across countries. For Germany, two main sources exist for vacancy data: the Federal Employment Agency (Bundesagentur) and the Job Vacancy Survey harmonised with EU regulations. The former is conducted monthly, while the latter is conducted quarterly. This study relies on quarterly data published by Destatis and reported by Eurostat, as these address the issue of underreported vacancies. The distinction between the two sources lies in the method of collection: monthly vacancies from Bundesagentur are based on firms voluntarily registering openings, whereas the quarterly survey involves active data collection. The same issue arises in the UK and Italy, where official agencies depend to some extent on voluntary firm reporting. As noted by the OECD (2017), employers frequently bypass official channels and rely on private recruitment methods, which compromises the reliability of administrative sources. In contrast, active sampling methods, which involve direct outreach to employers, improve coverage but are conducted less frequently. Methodological studies confirm that vacancy statistics from public employment services are inherently incomplete.

High-frequency data from online job postings offer an alternative way to measure labour demand. Covid provides a particularly relevant case, as remote work expanded rapidly due to lockdowns and subsequent adjustments in work practices, particularly in high-skilled occupations. Analyses of such data typically rely on language models trained to detect terms associated with remote work in job descriptions. Hansen and Lambert (2022) develop an algorithm that correctly classifies postings mentioning remote work in 99% of cases. Out-of-sample estimations based on this method show a large and persistent increase in demand for remote work from 2019 to 2022 — one of the lasting legacies of the pandemic. High-frequency data have also been employed to study labour reallocation during the crisis (Cœuré, 2020). In research on Beveridge curve shifts, such data are used as well: Benatti et al. (2020), for instance, document sharp declines in hiring between March and June 2020 by calculating the share of LinkedIn profiles that recorded a new employer in a given month. This methodology is promising for future work but remains outside the scope of this study. Its limitation lies in sectoral dependence, as platforms such as LinkedIn and Indeed primarily reflect the trend for higher-skilled, higher-paid jobs, leaving other segments of the labour market underrepresented.

Another approach to address the limitations of vacancy data is to substitute alternative indicators. Bonthuis et al. (2016), for instance, use an index of perceived labour shortages in manufacturing as a proxy for labour demand. Historical time series for labour shortages correlate well with official vacancy statistics, suggesting their potential usefulness. However, for the purposes of this study, such data are not available for separate countries across the entire period of interest. Moreover, manufacturing-specific shortages may not reflect overall labour demand after Covid as well as in the aftermath of the Great Financial Crisis. Consequently, this study remains with the general and transparent approach of measuring labour demand as the ratio of unfilled job positions to the total number of jobs.

4.2.2 Descriptive statistics

Stationarity tests indicate, as expected, that all series except the vacancy rates for UK are non-stationary. Visual inspection, as well as results obtained from previous research on the topic (Bonthuis et al., 2016; Kiss et al., 2022) suggest cointegration of the non-differenced series. Germany exhibits the lowest historical unemployment and highest vacancy rates, reflecting the efficiency of the country's labour market. UK's figures are similar, except both time series are more volatile. The worst performer is Italy, a result that

likely affected the results of the regression presented in the next section. While persistent unemployment that isn't very responsive to official vacancy developments is the main issue, the use of the vacancy series to characterise labour demand may also be problematic due to the nature of the Italian labour market, which features a high number of SMEs and where much if not most of hiring is done through unofficial channels, invisible to the vacancy statistic. This is in contrast to Germany and UK, where the market for jobs is more transparent, and where self-employment and part-time work do not play as big of a role (OECD, 2023b). Descriptive statistics for unemployment and JVR for the three countries are summarized in Table 1.

Table 1: Summary statistics for Germany, UK and Italy

	V_GE	U_GE	V_UK	U_UK	V_IT	U_IT
N	57	57	60	60	36	36
Mean	2.85	3.99	2.50	5.44	1.56	9.40
SD	0.70	0.87	0.67	1.62	0.54	1.72
Min	2	2.90	1.10	3.73	0.60	5.90
Max	4.50	6.30	4.10	8.39	2.30	11.90

4.3 Results

4.3.1 Germany

The results for Germany confirm the inverse relationship between unemployment and the job vacancy rate (JVR): the coefficient on the latter is negative and statistically significant at the 1% level. The first lag of unemployment is significant as well, indicating persistence in slack—a result commonly observed in European labour markets.

The Covid dummy variable is positive, suggesting an increase in unemployment following the outbreak, which corresponds to an outward shift of the Beveridge curve. Although the market trend has reversed since, the dummy captures the magnitude of the initial shock in 2020 and therefore remains positive.

As mentioned earlier, the first lag of the vacancy rate was included to improve regression accuracy. The model displays a very good fit, confirming the stability of the Beveridge curve for Germany in the period considered. This supports our earlier conclusion that the German labour market is characterised by strong co-movement of unemployment and vacancies.

Table 2: Beveridge curve estimation for Germany, 2010 Q4 to 2024 Q4. Data: *Eurostat*.

	<i>Dependent variable:</i>
	U_t
$L(U_t, 1)$	0.876*** (0.033)
V_t	-0.210*** (0.074)
$L(V_t, 1)$	0.085 (0.076)
D_C	0.092* (0.053)
Intercept	0.772*** (0.244)
Observations	56
R^2	0.972
Adjusted R^2	0.970
Residual Std. Error	0.141 (df = 51)
F Statistic	449.104*** (df = 4; 51)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

4.3.2 UK

For the UK, the Beveridge curve between 2010 and 2024 also holds well. The coefficient on the vacancy rate is negative and statistically significant, and the slope is similar to that observed in Germany. The lagged unemployment term is again positive and highly significant.

The dummy variable for the Covid period is positive, indicating more slack. This finding may appear contradictory, since the UK labour market tightened significantly in 2023–2024. However, the dummy is driven by the large outward shift in 2020, and the compensatory inward movement in later years is insufficient to outweigh it within the sample.

Overall, the model exhibits an excellent fit, with both series correlating strongly over time.

Table 3: Beveridge curve estimation for UK, 2010 Q1 to 2024 Q4.
Data: ONS.

	<i>Dependent variable:</i>
	U_t
V_t	-0.206*** (0.058)
$L(U_t, 1)$	0.932*** (0.025)
D_C	0.129* (0.070)
Intercept	0.786*** (0.254)
Observations	59
R^2	0.983
Adjusted R^2	0.983
Residual Std. Error	0.211 (df = 55)
F Statistic	1,089.393*** (df = 3; 55)

Note: *p<0.1; **p<0.05; ***p<0.01

4.3.3 Italy

Italy proves to be the most problematic case, as unemployment and vacancies show much weaker correlation. This aligns with earlier research, where identifying a Beveridge curve for Italy was empirically challenging (Bonthuis et al., 2013; Di Giorgio & Giannini, 2012).

The limited data sample (beginning only in 2016) reduces statistical power, and the results reflect this. The only regressor that is statistically significant is the first lag of unemployment. The inverse relationship between unemployment and vacancies is not confirmed, though differenced data indicates that unemployment and vacancy rates moved in opposite directions during the Covid crisis, as theory predicts.

Italy's particular labour market characteristics complicate econometric modelling. Persistent unemployment, the prevalence of SMEs, and reliance on informal hiring channels mean that official vacancy rates are less representative of true labour demand. As a result, while the outward shift of the curve is visible in descriptive analysis, the regression model does not yield robust evidence.

Table 4: Beveridge curve estimation for Italy, 2016 Q1 to 2024 Q4. Data: ISTAT.

	<i>Dependent variable:</i>
	U_t
V_t	0.036 (0.530)
$L(U_t, 1)$	0.702*** (0.137)
D_C	-0.417 (0.359)
Intercept	3.888** (1.774)
Observations	35
R^2	0.873
Adjusted R^2	0.857
Residual Std. Error	0.645 (df = 30)
F Statistic	51.767*** (df = 4; 30)
<i>Note:</i> *p<0.1; **p<0.05; ***p<0.01	

5 Conclusion

This paper has analysed the Beveridge curve in Germany, the UK, and Italy between 2010 and 2024, with particular emphasis on the Covid crisis. The results confirm that the fundamental inverse relationship between unemployment and vacancies holds in Germany and the UK, though with an outward shift during the pandemic. For Italy, the relationship is less clearly established, partly due to structural characteristics of its labour market and the limited availability of consistent vacancy data.

In Germany and the UK, unemployment persistence is strong, while vacancy rates respond more quickly to shocks. Both countries experienced an outward shift in 2020, reflecting the immediate dislocation caused by Covid. Germany's curve stabilised relatively quickly, aided by effective policy measures such as short-time work. The UK initially suffered a sharper shock but later recovered with higher vacancy rates, although the dummy variable still reflects the magnitude of the 2020 shift.

Italy's Beveridge curve is harder to model statistically, reflecting the mismatch between official vacancy data and actual hiring practices. While visual inspection suggests an outward shift, the regression results do not provide robust confirmation. This underlines the challenges of applying standard Beveridge curve analysis to economies with persistent structural unemployment and informal labour market dynamics.

Overall, the findings highlight the importance of labour market institutions and data quality in interpreting Beveridge curve shifts. The Covid crisis had a clear but temporary effect on European labour markets, with recovery trajectories shaped by country-specific conditions. Further research could expand the scope beyond these three countries, explore alternative measures of labour demand such as online job postings, and examine the role of inactivity and labour force participation more explicitly.

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